Question # 9 of 10 (Start time: 05:22:49 PM, 28 January 2023)

For a non empty closed and bounded subset S in $\mathbb{R}^n$, if H is the collection open sets such that $S \subset \cup\{H : H \in \mathrm{H}\}$, then by Heine - Borel theorem,

Select the correct option

- $S \subset \bigcup_{\lambda \in \Lambda} \{H_\lambda : H_\lambda \in \mathrm{H}\}$
- $S \subset \bigcup_{\alpha=1}^{\infty} \{H_\alpha : H_\alpha \in \mathrm{H}\}$
- All above are equally valid
- $S \subset \bigcup_{\alpha=1}^{n} \{H_\alpha : H_\alpha \in \mathrm{H}\}$

Click to Save Answer

MTH631 - Real Analysis II (Quiz 04)

Question # 4 of 10 (Start time: 05:20:24 PM, 28 January 2023)

An open disc: $\{(x, y) : x^2 + y^2 < 1\}$ in $\mathbb{R}^2$ is - - - - - - .

Select the correct option

○	diconnected
○	connected polygonally
○	none of these.
◉	disconnected polygonally

MTH631 - Real Analysis II (Quiz 04)

Question # 3 of 10 (Start time: 05:20:14 PM, 28 January 2023)

The set $\{(x, y) : -n < x, y < n, (x, y) \neq (0, 0), n \in \mathbb{N}\}$ is - - - - - - in $\mathbb{R}^2$.

Select the correct option

○	both open or closed
●	neither open nor closed
○	open
○	closed

MTH631 - Real Analysis II (Quiz 04)

Question # 1 of 10 (Start time: 05:18:58 PM, 28 January 2023

If $\phi \neq S \subseteq \mathbb{R}^n$, then the set S is bounded if - - - - - - - - .

Select the correct option

○	$\sup\{\|X-Y\| : X, Y \in S\} = \infty$
○	$\inf\{\|X-Y\| : X, Y \in S\} < \infty$
○	$\inf\{\|X-Y\| : X, Y \in S\} = \infty$
●	$\sup\{\|X-Y\| : X, Y \in S\} < \infty$

Question # 9 of 10 (Start time: 05:16:27 PM, 28 January 2023)

If $\mathbb{R}^n$ is connected, such that $\mathbb{R}^n = A \cup B$ with $\bar{A} \cap B = A \cap \bar{B} = \phi$, then - - - - - .

Select the correct option

- $\bar{A} \subset A, \bar{B} \subset B$
- Either $A = \mathbb{R}^n, B = \phi$ or $B = \mathbb{R}^n, A = \phi$
- All above can be concluded
- A and B are both open and closed

Click to Save Answer & N

MTH631 - Real Analysis II (Quiz 04)

Question # 8 of 10 (Start time: 05:16:15 PM, 28 January 2023)

If $\lim_{X \to X_0} f(X)$ exists, then it is _____.

Select the correct option

●	unique
○	1
○	none of these
○	0

Question # 7 of 10 (Start time: 05:15:18 PM, 28 January 2023

A compact set in $\mathbb{R}^n$ is - - - - - - - - - - .

Select the correct option

○	closed and unbounded
○	open and unbounded
○	open and bounded
●	closed and bounded

MTH631 - Real Analysis II (Quiz 04)

Question # 7 of 10 (Start time: 05:15:18 PM, 28 January 2023

A compact set in $\mathbb{R}^n$ is - - - - - - - - - - .

Select the correct option

○	closed and unbounded
○	open and unbounded
○	open and bounded
◉	closed and bounded

Question # 6 of 10 (Start time: 05:15:05 PM, 28 January 2023)

Intervals $(0,1)$ and $(1,2)$ are example of disconnected sets in $\mathbb{R}$ because - - - - - - - .

Select the correct option

- $\{\text{closure of } (0,1)\} \cap (1,2) = \phi$ and $\{\text{closure of } (1,2)\} \cap (0,1) = \phi$
- $(0,1) \cap (1,2) = \phi$

Click to Save Answer &

In $\mathbb{R}^n$, if $S_1, S_2, S_3 \ldots$ are non - empty closed subsets such that $S_1 \supset S_2 \supset \cdots \supset S_r \supset \cdots$ and $\sup\{|X_r - Y_r| : X_r, Y_r \in S_r, r \geqslant 1\} \to 0$ as $r \to \infty$,then the order of set $\bigcap_{r=1}^{\infty} S_r = - - - -$.

ect the correct option

Reload

- infinite
- one
- zero
- finite but more than one

Click to Save Answer & Move t

Question # 3 of 10 (Start time: 05:14:33 PM, 28 January 2023)

The set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ is - - - - - - in $\mathbb{R}^2$.

Select the correct option

◉	neither open nor closed
○	closed
○	open
○	both open or closed

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MTH631 - Real Analysis II (Quiz 04)

Question # 2 of 10 (Start time: 05:14:14 PM, 28 January 2023)

An open disc: $\{(x, y) : x^2 + y^2 < 1\}$ in $\mathbb{R}^2$ is - - - - - - - .

Select the correct option

○	diconnected
○	connected polygonally
◉	disconnected polygonally
○	none of these.

Question # 3 of 10 (Start time: 05:10:29 PM, 28 January 2023) Total

In $\mathbb{R}^n$, if $S_1, S_2, S_3 \ldots$ are non - empty closed subsets such that $S_1 \supset S_2 \supset \cdots \supset S_r \supset \cdots$ and $\sup\{|X_r - Y_r| : X_r, Y_r \in S_r, r \geqslant 1\} \to 0$ as $r \to \infty$,then the order of set $\bigcap_{r=1}^{\infty} S_r = ---$.

Select the correct option

Reload Math Equ

- infinite
- zero
- finite but more than one
- one

Click to Save Answer & Move to Next Que

Question # 2 of 10 (Start time: 05:10:10 PM, 28 January 2023) Total Mar

Set of isolated point(s) of the complement of set $\{(x,y) : -n < x, y < n, (x,y) \neq (0,0), n \in \mathbb{N}\}$ in $\mathbb{R}^2$, is $----$.

Select the correct option

Reload Math Equatio

- $\{(x,y) \in \mathbb{R}^2 : x = y = |n|, n \in \mathbb{N}\} \cap \{(0,0)\}$
- $\{(0,0)\}$
- $\{(x,y) \in \mathbb{R}^2 : x = y = |n|, n \in \mathbb{N}\} \cup \{(0,0)\}$
- $\{(x,y) \in \mathbb{R}^2 : x = y = |n|, n \in \mathbb{N}\}$

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 05:09:58 PM, 28 January 2023)

Let $f(x,y) = \frac{xy}{x^2+y^2}$ then the limit of f along the line $y = -x$ as (x,y) approach $(0,0)$ is _____.

Select the correct option

●	$-1/2$
○	$1/2$
○	undefined
○	0

Click to

Question # 10 of 10 (Start time: 05:08:16 PM, 28 January 2023)
Total M

In $\mathbb{R}^n$, if $S_1, S_2, S_3 \ldots$ are non - empty closed subsets such that $S_1 \supset S_2 \supset \cdots \supset S_r \supset \cdots$ and $\sup\{|X_r - Y_r| : X_r, Y_r \in S_r, r \geqslant 1\} \to 0$ as $r \to \infty$,then the order of set $\bigcap_{r=1}^{\infty} S_r = - - -$.

Select the correct option

Reload Math Equati

- one
- zero
- infinite
- finite but more than one

Question # 9 of 10 (Start time: 05:08:02 PM, 28 January 2023)

In $\mathbb{R}^3$, the lines $L_1 : X = (2,-1,5) + \alpha(2,-1,3)$ and $L_2 : X = (2,-1,5) + \beta\left(-5, \frac{5}{2}, -\frac{15}{2}\right)$ are traversed in - - - - - - - - - directions, where $-\infty < \alpha, \beta < \infty$.

Select the correct option

Relo

- opposite
- same
- oblique
- perpendicular

Click to Save Answer & Move

Question # 6 of 10 (Start time: 05:06:20 PM, 28 January

If $\lim_{X \to X_0} f(X)$ exists, then it is _____.

Select the correct option

○	0
◉	unique
○	1
○	none of these

Question # 2 of 10 (Start time: 05:02:02 PM, 28 January 2023) **Total Marks: 1**

The limit of the function $f(x) = \frac{xy}{x^2+y^2}$ by letting (x, y)approach $(0, 0)$along the line $y = x$ is _____.

Select the correct option

Reload Math Equations

○	Undefined
○	$-\frac{1}{2}$
◉	$\frac{1}{2}$
○	Finite

Click to Save Answer & Move to Next Question

Time Left sec(s)

MTH631 - Real Analysis II (Quiz 04)

Quiz Start Time: 05:01 PM, 28 January 2023

Question # 1 of 10 (Start time: 05:01:35 PM, 28 January 2023)

Total Marks: 1

Intervals $(0,1)$ and $(1,2)$ are example of disconnected sets in $\mathbb{R}$ because - - - - - - - .

Select the correct option

Reload Math Equations

- $(0,1) \cap (1,2) = \phi$
- $\{\text{closure of } (0,1)\} \cap (1,2) = \phi$ and $\{\text{closure of } (1,2)\} \cap (0,1) = \phi$

Click to Save Answer & Move to Next Question